

# STATRevision

## A Shiny Application for Revision Analysis

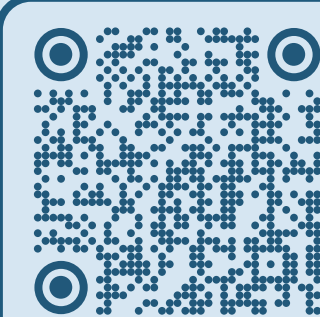
### Background

A **revision** is defined as the difference between a later result and a preliminary result

Distinction between **occasional** and **routine** revisions

**Routine revisions:** ongoing and based on a predefined publication procedure

**Example:** short-term business statistics (index of number of persons employed for industries and construction)



**2022 ESS peer review report for Austria:** development of the shiny application STATRevision in R by the Center for Methodology

### STATRevision: Development Goals



**User friendly:** Easy to use, mouseover tooltips, detailed explanations



**Standardisation** of revision analysis



**Peer Review improvement actions**



**Summary of results** should be exported directly from the application

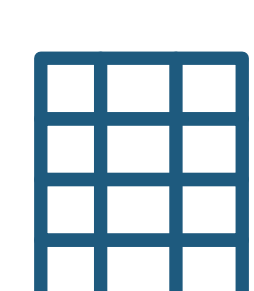
### Application Features



**Data transformation:** Adjust time span, select categories, calculate growth rates



**Visual Comparison:** of all vintages (release dates) or selected vintages



**Tables with statistical measures:** detailed results concerning the quality of revisions

### Conducting revision analysis with STATRevision

Shiny App is provided for staff at Statistics Austria and is locally accessible by Web-Browser

Main part consists of a dashboard with the following elements:

**A**

**Overall revision quality:** Table with key metrics for assessing different revisions

**B**

**Comparison (all estimators):** Graphical display of the results of all vintages

**C**

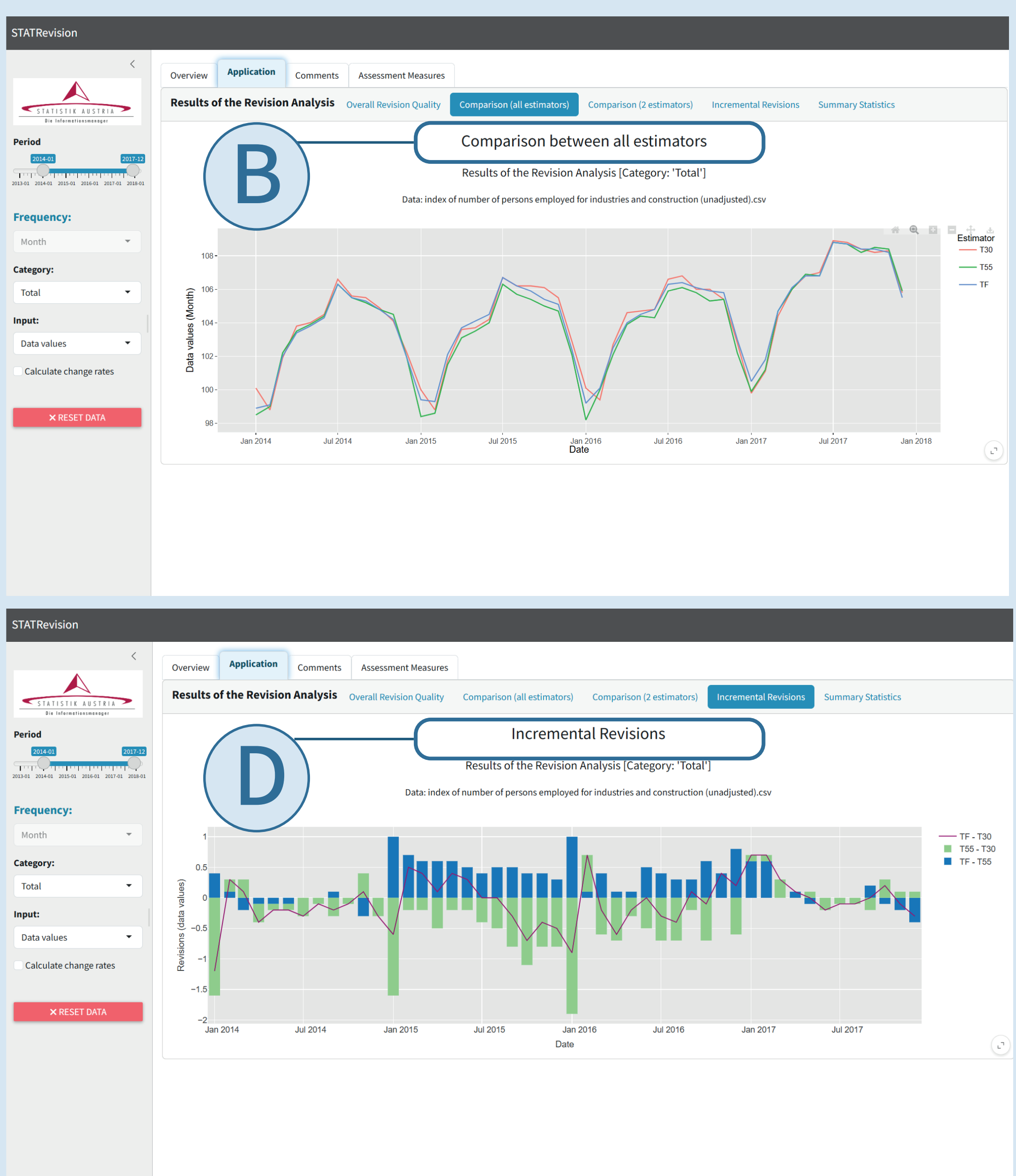
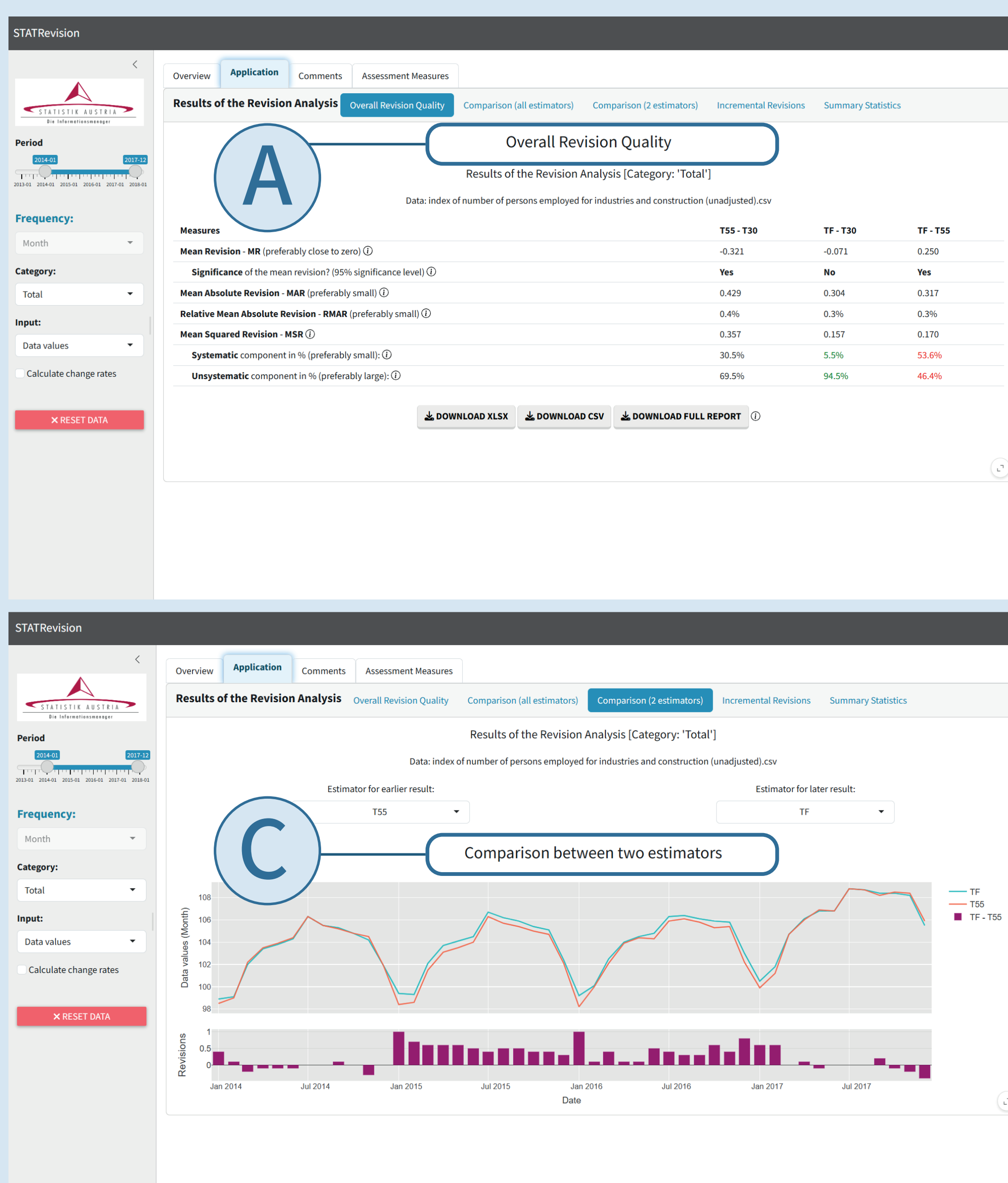
**Comparison (2 estimators):** Comparison chart between two user-selected results

**D**

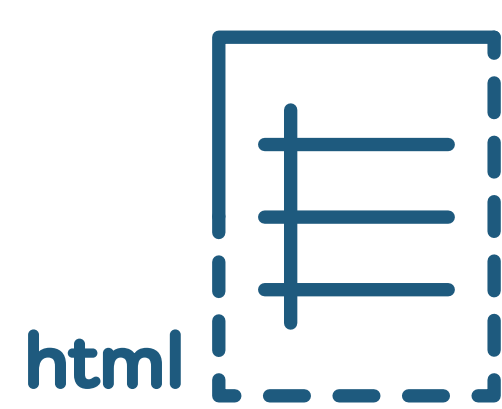
**Incremental revisions:** Chart showing revisions step by step across the sequence of individual vintages

#### Summary statistics:

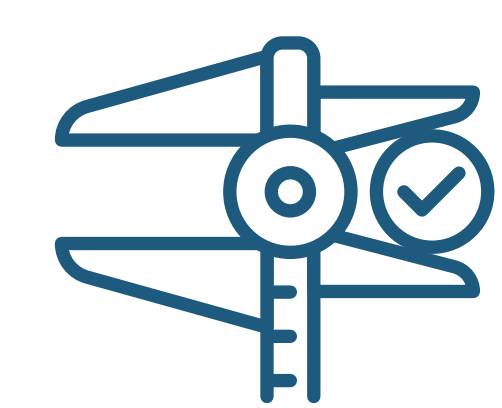
Table with an extended set of metrics for detailed revision analysis



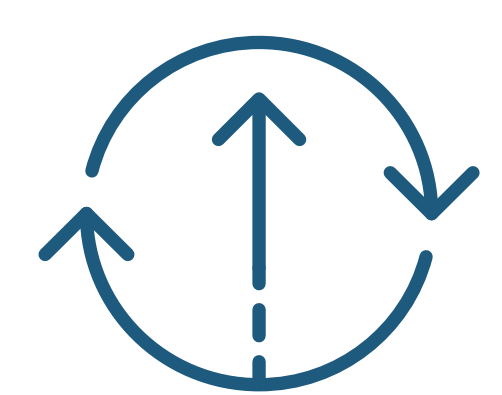
### Automated reporting



Subject matter project teams create **Revisions Analysis Report** in HTML format with the shiny app (Button "Download Full Report")



Center for Methodology **reviews results**



Automated upload of **overall revision quality results** to **intranet** site using Confluence API in R

### Further actions based on results of revision analysis

#### Concerns

If the mean revision differs significantly from zero, revisions are large on average, systematic component of the mean squared revision is substantial

#### Methodological Measures

Improving the method for early estimation, imputation in microdata, deeper analyses and evaluations of further methodological improvements

#### Improving Data Availability

Accelerating data collection, contacting data holders, examining alternative data sources